Statistical Computing

Constrained Optimization & Duality

Spring 2026

Outline

Constrained Optimization

Equality Constrained Optimization

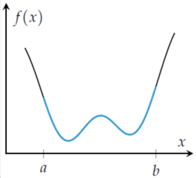
Inequality Constrained Optimization

Lagrangian Duality

Constrained Optimization

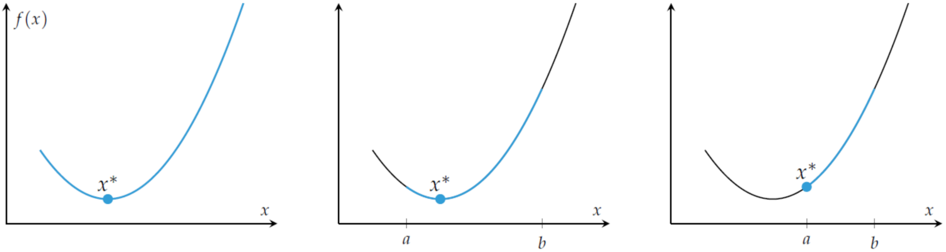
Constrained Maxima and Minima

- In many practical optimization problems, we must maximize or minimize a function in which the independent variables are subjected to certain further constraints.



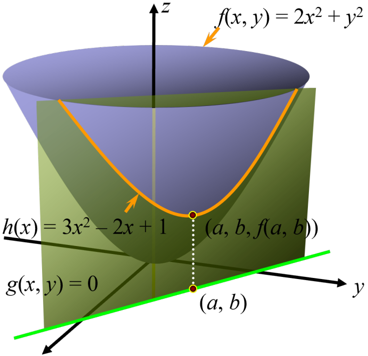
$$\begin{aligned} \underset{x}{\text{minimize}} \quad & f(x) \\ \text{subject to} \quad & x \in [a, b] \end{aligned}$$

Unconstrained   Constrained, Same Solution   Constrained, New Solution



Equality Constrained Optimization

- Find the relative minimum of $f(x, y) = 2x^2 + y^2$ subject to $g(x, y) = x + y - 1 = 0$.



Solution.

Solving the constraint for y gives $y = -x + 1$. Substituting into f:

$$h(x) = 2x^2 + (1 - x)^2 = 3x^2 - 2x + 1.$$

Solve $h'(x) = 6x - 2 = 0 \Rightarrow x = \frac{1}{3}$. Since $h''(x) = 6 > 0$, this is a relative minimum. With $y = \frac{2}{3}$, the constrained minimum is at $\left(\frac{1}{3}, \frac{2}{3}\right)$.

The Method of Lagrange Multipliers

- To find the relative extrema of $f(x, y)$ subject to $g(x, y) = 0$:

1. Form the **Lagrangian function** (where λ is the **Lagrange multiplier**):

$$F(x, y, \lambda) = f(x, y) + \lambda\, g(x, y).$$

2. Solve the system of equations

$$F_x = 0, \qquad F_y = 0, \qquad F_\lambda = 0$$

   for all values of x, y, and λ.

3. The solutions found in step 2 are candidates for the extrema of f.

Example: Lagrange Multipliers

- Find the relative minimum of $f(x,y) = 2x^2 + y^2$ subject to $x + y = 1$ using Lagrange multipliers.

Solution.

- Write $g(x,y) = x + y - 1 = 0$. Form the Lagrangian:

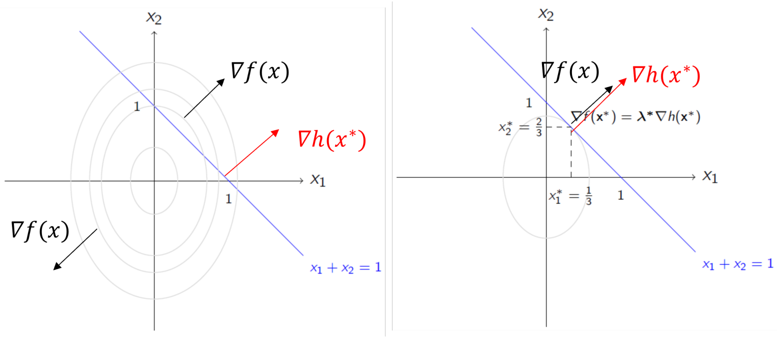
$$F(x,y,\lambda) = (2x^2 + y^2) + \lambda(x + y - 1).$$

- Solve the system:

$$F_x = 4x + \lambda = 0, \qquad F_y = 2y + \lambda = 0, \qquad F_\lambda = x + y - 1 = 0.$$

- From the first two equations: $x = -\frac{\lambda}{4}$ and $y = -\frac{\lambda}{2}$.
- Substituting into the third: $\lambda = -\frac{4}{3}$, so $x = \frac{1}{3}$, $y = \frac{2}{3}$.

$$\begin{array}{ll} \text{Minimize} & 2x_1^2 + x_2^2 \\ \text{Subject to} & x_1 + x_2 = 1 \end{array}$$



Geometric Interpretation

- Consider the gradients of f and h at the optimal point $\boldsymbol{x}^*$.
- They must be parallel (same direction, possibly different lengths):

$$\nabla f(\boldsymbol{x}^*) = \nu^* \, \nabla h(\boldsymbol{x}^*).$$

- This is equivalent to $\nabla L(\boldsymbol{x}^*, \nu^*) = 0$ along with feasibility of $\boldsymbol{x}^*$.
- $\nabla f(\boldsymbol{x})$ is perpendicular to a level curve of f; similarly $\nabla h(\boldsymbol{x})$ is perpendicular to the level curve $h(\boldsymbol{x}) = 0$.
- From the example, at $x_1^* = \frac{1}{3}$, $x_2^* = \frac{2}{3}$, $\lambda_1^* = \frac{4}{3}$:

$$\nabla f(x_1^*, x_2^*) = \begin{bmatrix} \frac{4}{3} & \frac{4}{3} \end{bmatrix}, \qquad \nabla h_1(x_1^*, x_2^*) = \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

Equality Constrained Optimization

- Optimization problem with one equality constraint:

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & h(\boldsymbol{x}) = 0. \end{aligned}$$

- The **Lagrangian** is $L(\boldsymbol{x}, \nu) = f(\boldsymbol{x}) + \nu h(\boldsymbol{x})$.
- A solution must satisfy $\nabla_{\boldsymbol{x}} L(\boldsymbol{x}^*, \nu^*) = \nabla f(\boldsymbol{x}^*) + \nu^* \nabla h(\boldsymbol{x}^*) = \mathbf{0}$, i.e., there exists ν^* such that

$$\frac{\partial L(\boldsymbol{x}^*, \nu^*)}{\partial x_1} = \cdots = \frac{\partial L(\boldsymbol{x}^*, \nu^*)}{\partial x_n} = 0, \qquad \frac{\partial L(\boldsymbol{x}^*, \nu^*)}{\partial \nu} = h(\boldsymbol{x}^*) = 0.$$

Equality Constrained Optimization: Two Constraints

- Optimization problem with two equality constraints:

$$\begin{array}{rl} \text{minimize} & f(\boldsymbol{x}) \\ \text{subject to} & h_1(\boldsymbol{x}) = 0, \; h_2(\boldsymbol{x}) = 0. \end{array}$$

- The Lagrangian is $L(\boldsymbol{x}, \boldsymbol{\nu}) = f(\boldsymbol{x}) + \nu_1 h_1(\boldsymbol{x}) + \nu_2 h_2(\boldsymbol{x})$.
- Stationarity condition:

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\nu}) = \nabla f(\boldsymbol{x}) + \nu_1 \nabla h_1(\boldsymbol{x}) + \nu_2 \nabla h_2(\boldsymbol{x}) = \mathbf{0}.$$

Equality Constrained Optimization: General Case

- General problem with m equality constraints:

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & h_j(\boldsymbol{x}) = 0, \quad j = 1, \ldots, m. \end{aligned}$$

- Denote the constraint vector and multiplier vector:

$$\boldsymbol{h}(\boldsymbol{x}) = \begin{pmatrix} h_1(\boldsymbol{x}) \\ \vdots \\ h_m(\boldsymbol{x}) \end{pmatrix} \in \mathbb{R}^m, \qquad \boldsymbol{\nu} = \begin{pmatrix} \nu_1 \\ \vdots \\ \nu_m \end{pmatrix} \in \mathbb{R}^m.$$

- The Lagrangian is

$$L(\boldsymbol{x}, \boldsymbol{\nu}) = f(\boldsymbol{x}) + \sum_{j=1}^{m} \nu_j h_j(\boldsymbol{x}) = f(\boldsymbol{x}) + \boldsymbol{\nu}^\top \boldsymbol{h}(\boldsymbol{x}).$$

Equality Constrained Optimization: General Case (cont.)

- **First-Order Necessary Conditions (equality constraints only):**
- Suppose $\boldsymbol{x}^*$ is a local solution and LICQ holds at $\boldsymbol{x}^*$.
  - **LICQ** (Linear Independence Constraint Qualification): the constraint gradients $\nabla h_1(\boldsymbol{x}^*), \ldots, \nabla h_m(\boldsymbol{x}^*)$ are linearly independent.
- Then there exists $\boldsymbol{\nu}^*$ such that at $(\boldsymbol{x}^*, \boldsymbol{\nu}^*)$:

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}^*, \boldsymbol{\nu}^*) = \nabla f(\boldsymbol{x}^*) + \sum_{j=1}^{m} \nu_j^* \nabla h_j(\boldsymbol{x}^*) = \mathbf{0},$$

$$h_j(\boldsymbol{x}^*) = 0, \qquad j = 1, \ldots, m.$$

LICQ: Why Linear Independence?

- **Example:**

$$\begin{aligned} \text{Minimize} \quad & x_1 + x_2 + x_3^2 \\ \text{Subject to:} \quad & x_1 = 1, \quad x_1^2 + x_2^2 = 1. \end{aligned}$$

The minimum is achieved at $x_1 = 1$, $x_2 = 0$, $x_3 = 0$.

The Lagrangian is:

$$L(x_1, x_2, x_3, \lambda_1, \lambda_2) = x_1 + x_2 + x_3^2 + \lambda_1(1 - x_1) + \lambda_2(1 - x_1^2 - x_2^2).$$

But $\dfrac{\partial L(1, 0, 0, \lambda_1, \lambda_2)}{\partial x_2} = 1$ for all λ_1, λ_2: no multipliers can satisfy stationarity!

Observe: $\nabla h_1(1, 0, 0) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$ and $\nabla h_2(1, 0, 0) = \begin{bmatrix} 2 & 0 & 0 \end{bmatrix}$ are *linearly dependent*.

Inequality Constrained Optimization

Inequality Constrained Optimization

- Optimization problem with one inequality constraint:

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g(\boldsymbol{x}) \leq 0. \end{aligned}$$

- Two cases arise depending on where the constrained minimum $\boldsymbol{x}^*$ lies:
  - $g(\boldsymbol{x}^*) = 0$ (boundary): the constraint is active; the solution is pushed onto the constraint boundary.
  - $g(\boldsymbol{x}^*) < 0$ (interior): the constraint is inactive; the solution coincides with the unconstrained minimum and $\lambda^* = 0$.

- If the solution lies on the boundary ($g(\boldsymbol{x}^*) = 0$), the constraint is active and behaves like an equality constraint. The same Lagrange condition applies:

$$\nabla f(\boldsymbol{x}^*) + \lambda^* \nabla g(\boldsymbol{x}^*) = \mathbf{0}.$$

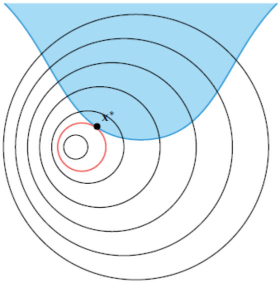


Figure 10.5. An active inequality constraint. The corresponding contour line is shown in red.

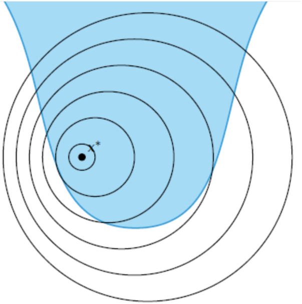


Figure 10.6. An inactive inequality constraint.

Example: Inequality Constraint

$$\begin{aligned} \text{Minimize} \quad & 3x_1^2 + x_2^2 \\ \text{subject to} \quad & x_1 + x_2 \geq 1 \end{aligned}$$

The unconstrained minimum $(0, 0)$ violates $x_1 + x_2 \geq 1$, so the constraint is active at $\boldsymbol{x}^*$.

With $g(\boldsymbol{x}) = 1 - x_1 - x_2$, stationarity of $L = 3x_1^2 + x_2^2 + \lambda(1 - x_1 - x_2)$ gives $x_1 = \frac{\lambda}{6}$, $x_2 = \frac{\lambda}{2}$. Substituting into the constraint:

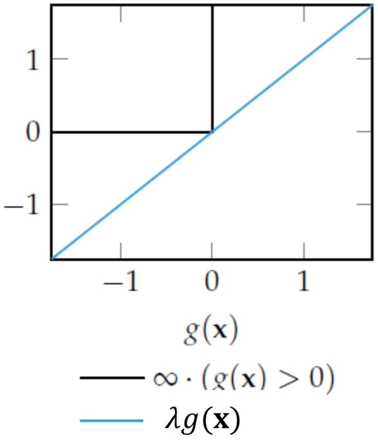
$$\frac{\lambda}{6} + \frac{\lambda}{2} = 1 \;\Rightarrow\; \lambda^* = \frac{3}{2}, \qquad \boldsymbol{x}^* = \left(\frac{1}{4}, \frac{3}{4}\right).$$

Motivation for the KKT (Karush-Kuhn-Tucker) Conditions

- Imposing $g(\boldsymbol{x}) \leq 0$ is equivalent to minimizing a step-penalized objective:

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) \text{ s.t. } g(\boldsymbol{x}) \leq 0 \iff \min_{\boldsymbol{x}} f_\infty(\boldsymbol{x}),$$

  where $f_\infty(\boldsymbol{x}) := f(\boldsymbol{x}) + \infty \cdot \mathbf{1}_{g(\boldsymbol{x})>0}(\boldsymbol{x})$ equals $f(\boldsymbol{x})$ when $g(\boldsymbol{x}) \leq 0$ and $+\infty$ otherwise.
- We use a linear penalty $\lambda g(\boldsymbol{x})$ ($\lambda \geq 0$) as a lower bound on $\infty \cdot \mathbf{1}_{g(\boldsymbol{x})>0}(\boldsymbol{x})$, penalizing infeasibility.



Motivation for the KKT Conditions

- **Lagrangian for inequality constraints:**

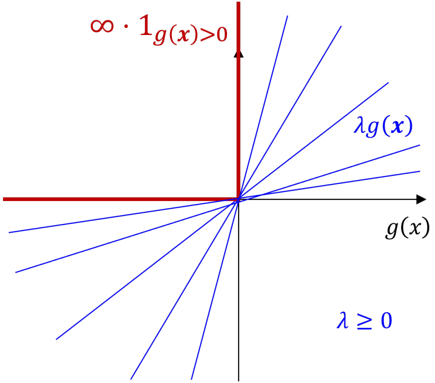
$$L(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) + \lambda g(\boldsymbol{x}).$$

- Recover f_∞ by maximizing over λ:

$$f_\infty(\boldsymbol{x}) = \max_{\lambda \geq 0} L(\boldsymbol{x}, \lambda).$$

- **(Primal problem)**

$$\min_{\boldsymbol{x}} \max_{\lambda \geq 0} L(\boldsymbol{x}, \lambda).$$



KKT Conditions

From $\min_{\boldsymbol{x}} \max_{\lambda \geq 0} L(\boldsymbol{x}, \lambda) := f(\boldsymbol{x}) + \lambda g(\boldsymbol{x})$, the following conditions must hold at $\boldsymbol{x}^*$:

- **Primal feasibility:** $g(\boldsymbol{x}^*) \leq 0$.
- **Dual feasibility:** $\lambda^* \geq 0$, required for $\lambda g(\boldsymbol{x})$ to serve as a lower bound on the step penalty.
- **Complementary slackness:** $\lambda^* g(\boldsymbol{x}^*) = 0$.
  - When $g(\boldsymbol{x}^*) < 0$, λ^* must be 0 to recover $f(\boldsymbol{x}^*)$ from $\max_{\lambda \geq 0} L(\boldsymbol{x}^*, \lambda)$.
  - When $g(\boldsymbol{x}^*) = 0$, λ^* may be positive.
- **Stationarity:** $\nabla_{\boldsymbol{x}} L(\boldsymbol{x}^*, \lambda^*) = \nabla f(\boldsymbol{x}^*) + \lambda^* \nabla g(\boldsymbol{x}^*) = \mathbf{0}$.
  - When the constraint is active, the level curves of f and g must be aligned.
  - When inactive, $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ and $\lambda^* = 0$.

KKT Conditions: General Case

- General problem with l inequality constraints:

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \leq 0, \quad i = 1, \ldots, l. \end{aligned}$$

- $L(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \sum_{i=1}^{l} \lambda_i g_i(\boldsymbol{x})$
- The active set at $\boldsymbol{x}^*$ is $\mathcal{A}(\boldsymbol{x}^*) = \{i : g_i(\boldsymbol{x}^*) = 0\}$. **LICQ** holds if the gradients $\{\nabla g_i(\boldsymbol{x}^*) : i \in \mathcal{A}(\boldsymbol{x}^*)\}$ are linearly independent.
- If $\boldsymbol{x}^*$ is a local solution and LICQ holds, there exist λ_i^* satisfying **KKT conditions**:

$$\nabla_{\boldsymbol{x}} L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*) = \nabla f(\boldsymbol{x}^*) + \sum_{i=1}^{l} \lambda_i^* \nabla g_i(\boldsymbol{x}^*) = \mathbf{0},$$

$$g_i(\boldsymbol{x}^*) \leq 0, \quad \lambda_i^* \geq 0, \quad \lambda_i^* g_i(\boldsymbol{x}^*) = 0, \qquad i = 1, \ldots, l.$$

Example: KKT with Inequality Constraints

$$\begin{array}{ll} \text{maximize} & -(x-2)^2 - 2(y-1)^2 \\ \text{subject to} & x + 4y \leq 3, \quad x \geq y \end{array} \Rightarrow \begin{array}{ll} \text{minimize} & (x-2)^2 + 2(y-1)^2 \\ \text{subject to} & x + 4y - 3 \leq 0, \quad -x + y \leq 0. \end{array}$$

Lagrangian: $L = (x-2)^2 + 2(y-1)^2 + \mu_1(x + 4y - 3) + \mu_2(-x + y)$.

KKT stationarity:

$$\frac{\partial L}{\partial x} = 2(x-2) + \mu_1 - \mu_2 = 0, \qquad \frac{\partial L}{\partial y} = 4(y-1) + 4\mu_1 + \mu_2 = 0.$$

Complementary slackness: $\mu_1(x + 4y - 3) = 0, \quad \mu_2(-x + y) = 0, \quad \mu_1, \mu_2 \geq 0.$

Example: KKT with Inequality Constraints (continued)

Two complementarity conditions $\Rightarrow$ check 4 cases:

- **Case 1** $\mu_1 = \mu_2 = 0$: stationarity $\Rightarrow x = 2$, $y = 1$. But $2 + 4(1) = 6 > 3$: *infeasible*.
- **Case 2** $\mu_1 = 0$, $x = y$: stationarity $\Rightarrow x = y = \frac{4}{3}$, but $\mu_2 = -\frac{4}{3} < 0$: *dual infeasible*.
- **Case 3** $x + 4y = 3$, $\mu_2 = 0$: stationarity $\Rightarrow x = \frac{5}{3}$, $y = \frac{1}{3}$, $\mu_1 = \frac{2}{3} > 0$.
  Inactive constraint: $-x + y = -\frac{4}{3} \leq 0$ ✓. **Valid KKT point.**
- **Case 4** $x + 4y = 3$, $x = y$: stationarity $\Rightarrow x = y = \frac{3}{5}$, but $\mu_2 = -\frac{48}{25} < 0$: *dual infeasible*.

Optimal solution: $x^* = \frac{5}{3}$, $y^* = \frac{1}{3}$, $(x^* - 2)^2 + 2(y^* - 1)^2 = 1$.

Example: KKT Conditions

$$\begin{array}{rl} \text{maximize} & -(x-3)^2-(y-2)^2 \\ \text{subject to} & x^2+y^2 \leq 5, \quad x+2y \leq 4, \quad x, y \geq 0 \end{array}$$

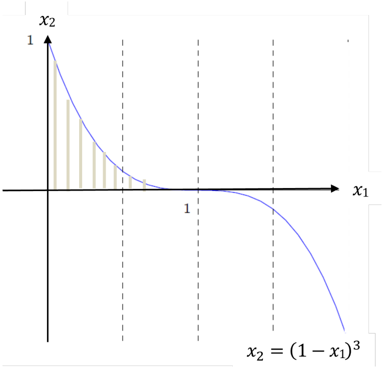
$$\nabla f = (6-2x, 4-2y),\ \nabla g_1 = (2x, 2y),\ \nabla g_2 = (1,2),\ \nabla g_3 = (-1,0),\ \nabla g_4 = (0,-1).$$

At $(x, y) = (2, 1)$: feasible with active set $\mathcal{I} = \{1, 2\}$. KKT stationarity:

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} - \mu_1 \begin{bmatrix} 4 \\ 2 \end{bmatrix} - \mu_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \mathbf{0} \quad \Rightarrow \quad \mu_1 = \frac{1}{3}, \quad \mu_2 = \frac{2}{3}.$$

Example: KKT with Multiple Constraints

$$
\begin{aligned}
\text{minimize} \quad & \left(x_1 - \tfrac{3}{2}\right)^2 + \left(x_2 - \tfrac{1}{2}\right)^2 \\
\text{subject to} \quad & c_1(\boldsymbol{x}) = x_1 + x_2 - 1 \leq 0, \\
& c_2(\boldsymbol{x}) = x_1 - x_2 - 1 \leq 0, \\
& c_3(\boldsymbol{x}) = -x_1 + x_2 - 1 \leq 0, \\
& c_4(\boldsymbol{x}) = -x_1 - x_2 - 1 \leq 0.
\end{aligned}
$$



- Solution: $\boldsymbol{x}^* = (1, 0)^\mathsf{T}$; active constraints: c_1, c_2.

$$
\nabla f(\boldsymbol{x}^*) = \begin{bmatrix} -1 \\ -\frac{1}{2} \end{bmatrix}, \ \nabla c_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \ \nabla c_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.
$$

- KKT multipliers: $\boldsymbol{\lambda}^* = \left(\frac{3}{4}, \frac{1}{4}, 0, 0\right)^\mathsf{T}$.

Exercise

- **Class Exercise 1**

$$\begin{aligned} \text{minimize} \quad & 2x_1^2 + 2x_1x_2 + x_2^2 - 10x_1 - 10x_2 \\ \text{subject to} \quad & x_1^2 + x_2^2 \le 5, \quad 3x_1 + x_2 \le 6. \end{aligned}$$

Solution: $x_1^* = 1,\ x_2^* = 2,\ \lambda^* = 1.$

- **Class Exercise 2**

$$\begin{aligned} \text{maximize} \quad & \ln x + 1 + y \\ \text{subject to} \quad & 2x + y \le 3, \quad x, y \ge 0. \end{aligned}$$

- **Class Exercise 3**

$$\begin{aligned} \text{minimize} \quad & x^2 + y^2 \\ \text{subject to} \quad & x^2 + y^2 \le 5, \quad x + 2y = 4, \quad x, y \ge 0. \end{aligned}$$

Lagrangian Duality

Motivation for Duality

Primal problem

$$\min_{\boldsymbol{x}} \max_{\boldsymbol{\lambda} \geq 0, \boldsymbol{\nu}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$$

Dual problem

$$\max_{\boldsymbol{\lambda} \geq 0, \boldsymbol{\nu}} \min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$$

- For any $(\boldsymbol{\lambda}, \boldsymbol{\nu})$:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq \max_{\boldsymbol{\lambda}' \geq 0, \boldsymbol{\nu}'} L(\boldsymbol{x}, \boldsymbol{\lambda}', \boldsymbol{\nu}').$$

- Therefore:

$$\min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq \min_{\boldsymbol{x}} \max_{\boldsymbol{\lambda}' \geq 0, \boldsymbol{\nu}'} L(\boldsymbol{x}, \boldsymbol{\lambda}', \boldsymbol{\nu}').$$

- Hence (**weak duality**):

$$\max_{\boldsymbol{\lambda} \geq 0, \boldsymbol{\nu}} \min_{\boldsymbol{x}} L \leq \min_{\boldsymbol{x}} \max_{\boldsymbol{\lambda} \geq 0, \boldsymbol{\nu}} L.$$

Motivation for Duality (continued)

- The solution to the dual problem provides a **lower bound** to the primal optimum.

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^\top \boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{\nu}^\top \boldsymbol{h}(\boldsymbol{x}).$$

$$\begin{aligned} &\text{minimize } f(\boldsymbol{x}) \\ &\text{subject to } \boldsymbol{g}(\boldsymbol{x}) \leq 0, \quad \boldsymbol{h}(\boldsymbol{x}) = 0. \end{aligned}$$

The Lagrangian Duality

- **Primal problem (P):**

$$
\begin{aligned}
\text{minimize } & f(x) \\
\text{subject to } & g_i(x) \le 0, \quad i = 1, \ldots, l, \\
& h_j(x) = 0, \quad j = 1, \ldots, m.
\end{aligned}
$$

- **Lagrangian Dual Problem (D):**

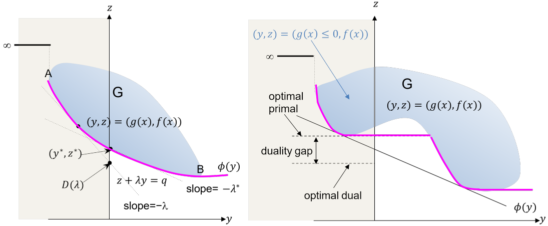
$$
\begin{aligned}
\text{maximize } & \mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}) \triangleq \min_x \ L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \\
\text{subject to } & \lambda_i \ge 0, \quad i = 1, \ldots, l,
\end{aligned}
$$

where

$$
L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\boldsymbol{x}) + \sum_i \lambda_i g_i(\boldsymbol{x}) + \sum_j \nu_j h_j(\boldsymbol{x}).
$$

(*) Geometric Interpretation of Lagrangian Duality

- Consider the primal problem with one inequality constraint: $\min f(x)$ s.t. $g(x) \le 0$.
- In the (y,z)-plane, let $G = \{(y,z) : y = g(x),\ z = f(x)\}$. The primal problem finds a point in G with $y \le 0$ that minimizes z.
- To compute $\mathcal{D}(\lambda)$ for $\lambda \ge 0$: minimize $z + \lambda y = f(x) + \lambda g(x)$ over points in G.
- The dual problem is equivalent to finding the slope λ of the supporting hyperplane to G whose z-intercept is maximal.



Weak Duality Theorem

Let $\boldsymbol{x}$ be feasible for (P) and $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ be feasible for (D) (i.e., $\boldsymbol{\lambda} \geq 0$). Then

$$f(\boldsymbol{x}) \geq \mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

Moreover, if $f(\boldsymbol{x}^*) = \mathcal{D}(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$, then $\boldsymbol{x}^*$ and $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ are optimal for (P) and (D) respectively.

Proof. Since $\mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_y L(y, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ and $\boldsymbol{\lambda} \geq \mathbf{0}$:

$$\mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f(\boldsymbol{x}) + \underbrace{\boldsymbol{\lambda}^{\mathsf{T}} \boldsymbol{g}(\boldsymbol{x})}_{\leq 0} + \underbrace{\boldsymbol{\nu}^{\mathsf{T}} \boldsymbol{h}(\boldsymbol{x})}_{= 0} \leq f(\boldsymbol{x}). \quad \square$$

Duality Gap

Duality Gap

If the optimal objective value of the primal problem is *strictly greater* than the optimal objective value of the dual problem, a **duality gap** is said to exist:

$$f^* - \mathcal{D}^* > 0.$$

Strong Duality Theorem

Let f, g_i be convex ($i = 1, \ldots, l$) and h_j be affine ($j = 1, \ldots, m$). Suppose **Slater's condition** holds: $\exists\, x \in \mathbb{R}^n$ such that $g_i(x) < 0$ and $h_j(x) = 0$. Then

$$\inf\{f(x) : g_i(x) \leq 0,\ h_j(x) = 0\} = \sup\{\mathcal{D}(\boldsymbol{\lambda}, \boldsymbol{\nu}) : \boldsymbol{\lambda} \geq 0\}.$$

If the inf is finite, the sup is achieved at some $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ with $\boldsymbol{\lambda}^* \geq 0$. If the inf is achieved at x^*, then $\lambda_i^* g_i(x^*) = 0$ for all i.

- The Lagrangian $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ is convex in $\boldsymbol{x}$ under these conditions.

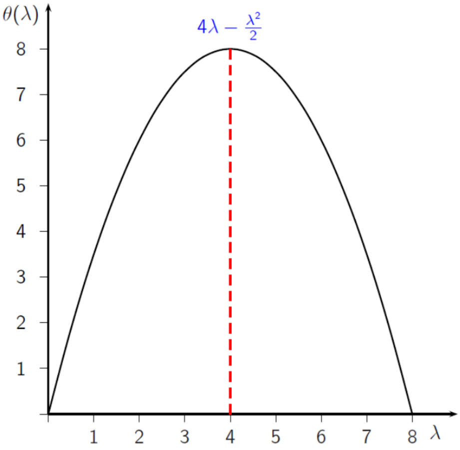
Example: Strong Duality

- Consider:

$$\text{minimize } x_1^2 + x_2^2$$
$$\text{subject to } x_1 + x_2 \geq 4, \quad x_1, x_2 \geq 0.$$

Let $X = \mathbb{R}^2_+$. Lagrangian:
$L(x, \lambda) = x_1^2 + x_2^2 + \lambda(4 - x_1 - x_2)$.
Dual function:

$$\theta(\lambda) = 4\lambda + \min_{x_1 \geq 0}\{x_1^2 - \lambda x_1\} + \min_{x_2 \geq 0}\{x_2^2 - \lambda x_2\}.$$



Graph of the dual function $\theta(\lambda)$

Example: Strong Duality (continued)

The minimum of $L(x, \lambda)$ over $X = \mathbb{R}^2_+$ is attained at

$$x_1(\lambda) = \frac{\lambda}{2}, \qquad x_2(\lambda) = \frac{\lambda}{2}.$$

Thus:

$$\theta(\lambda) = L(x(\lambda), \lambda) = 4\lambda - \frac{\lambda^2}{2} \quad \forall\, \lambda \geq 0.$$

The dual function is concave and differentiable. Maximize:

$$\frac{\partial \theta}{\partial \lambda} = 4 - \lambda = 0 \;\Rightarrow\; \lambda^* = 4.$$

Therefore $\theta(\lambda^*) = 8$ and $x^* = x(\lambda^*) = (2, 2)$. No duality gap: $f(x^*) = 8 = \theta(\lambda^*)$. ✓

Exercise

- **Class Exercise**
  - Solve the problem.
  - Formulate and solve the dual problem.
  - Check whether the primal and dual objective values are equal (strong duality).

$$\begin{aligned} &\text{minimize } x_1 \\ &\text{subject to } x_1^2 + x_2^2 = 1. \end{aligned}$$

Duality in Convex Problems: Linear Programming

- **Primal LP:**

$$\text{minimize } c^\top x$$

$$\text{subject to } Ax = b, \quad x \geq 0.$$

- Lagrangian: $L(x, \boldsymbol{\lambda}, \boldsymbol{\nu}) = c^\top x - \boldsymbol{\lambda}^\top x + \boldsymbol{\nu}^\top (b - Ax)$.
- Stationarity for the dual: $c - \boldsymbol{\lambda} - A^\top \boldsymbol{\nu} = 0$, $\boldsymbol{\lambda} \geq 0$.
- **Dual LP:**

$$\text{maximize } b^\top \boldsymbol{\nu}$$

$$\text{subject to } A^\top \boldsymbol{\nu} \leq c.$$

Duality in Convex Problems: Quadratic Programming

- **Primal QP** ($H \succeq 0$):

$$\begin{aligned} \text{minimize } & \tfrac{1}{2}x^{\mathsf{T}}Hx + d^{\mathsf{T}}x \\ \text{subject to } & Ax \le b. \end{aligned}$$

- Lagrangian: $L(x, \boldsymbol{\lambda}) = \frac{1}{2}x^{\mathsf{T}}Hx + d^{\mathsf{T}}x + \boldsymbol{\lambda}^{\mathsf{T}}(Ax - b)$.
- Stationarity: $Hx + A^{\mathsf{T}}\boldsymbol{\lambda} + d = 0 \;\Rightarrow\; x = -H^{-1}(d + A^{\mathsf{T}}\boldsymbol{\lambda})$.
- **Dual QP:**

$$\begin{aligned} \text{maximize } & \tfrac{1}{2}\boldsymbol{\lambda}^{\mathsf{T}}D\boldsymbol{\lambda} + \boldsymbol{\lambda}^{\mathsf{T}}c - \tfrac{1}{2}d^{\mathsf{T}}H^{-1}d \\ \text{subject to } & \boldsymbol{\lambda} \ge 0, \end{aligned}$$

  where $D = -AH^{-1}A^{\mathsf{T}}$ and $c = -b - AH^{-1}d$.
  - This maximizes a *concave* quadratic over the nonnegative orthant.